

Executive Summary

Project Title

Predictive Modeling and Risk Scoring for Bank Customer Churn

Overview

This project developed an AI-powered machine learning system capable of predicting customer churn and generating probability-based risk scores for retail banking environments.

The solution was designed to help banks identify customers who are likely to leave the organization before churn occurs, enabling proactive retention strategies and data-driven customer engagement.

Problem Addressed

Traditional banking churn analysis methods are often reactive and unable to identify high-risk customers early.

The project addressed the following challenges:

- Lack of predictive churn intelligence
 - Absence of customer-level risk scoring
 - Limited explainability of churn behavior
 - Inefficient retention planning
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Solution Developed

The project implemented:

- Machine learning-based churn prediction
- Customer risk scoring framework
- Explainable AI using SHAP
- Interactive Streamlit dashboard

The system predicts customer churn probabilities and categorizes customers into:

- Low Risk
 - Medium Risk
 - High Risk
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Technologies Used

- Python
- Pandas
- Scikit-learn
- SHAP

- Streamlit
 - Matplotlib
 - Seaborn
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Key Outcomes

- Achieved strong predictive performance using Random Forest Classifier
 - Generated explainable churn insights using SHAP
 - Developed a deployed AI-powered Streamlit dashboard
 - Created an interactive customer risk analysis system
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Business Impact

The developed solution can help banks:

- Improve customer retention strategies
 - Identify high-risk customers proactively
 - Enhance customer engagement planning
 - Support data-driven business decision-making
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Conclusion

This project successfully transformed traditional churn analysis into an intelligent predictive analytics solution by combining machine learning, explainable AI, and interactive deployment technologies.

The system demonstrates the practical application of AI in solving real-world banking and customer retention challenges.